Ch#3 "Images"

#What is still image? and how it is created?

- Still images may be the most important element of a multimedia project
- Still images are visual representations that do not move.
- A still image is data in which a grid or raster of picture elements pixels has been mapped to represent a visual subject.
  e.g the page of a book or a photograph.

- The type of still images created depends on the display resolution, and hardware and software capabilities.

Types of still images:
- Rendering
- 3-D drawing

- Still images are generated in two ways:
  - Bitmaps
  - vector drawn graphics.

What is bitmap image?

=> Bitmap is derived from the words "bit", which means the simplest element in which only two digits are used,
=> And "map" which is a two dimensional matrix of these bits.
=> A bitmap is a data matrix describing the individual dots of an image.
=> Also called "raster images".

What is the usage of bitmap image format?

- Bitmaps are an image format suited for creation of:
  - Photo-realistic images
  - Complex drawings
  - Images that require fine detail.
- Bitmapped images are known as paint graphics.
- A bitmap is made up of individual dots or picture elements known as pixels or pels.
- Bitmapped images can have varying bit & color depths.

Difference b/w painting program and drawing program?

Painting Program	Drawing Program
○ Bitmap editors are called painting program.	○ And vector editors are called drawing program.

Why image files are compressed

=> Bitmap images and vector images are stored in various file formats and can be translated from one application to another or from one computer platform to another.

=> Typically image files are compressed:
- To save memory
- To save disk space.

=> many bitmap images file formats already use compression within the file itself.

for example:
- GIF
- JPEG
- PNG

What is image?

○ Images (Bitmap/Raster) are defined as matrices where each element represents a pixel (picture element).

○ Each pixel represents color information for a specific location in the image.

○ There can be only one color value per pixel.

○ There are three primary colors that represents all colors:
- ○ Red
- ○ Green
- ○ Blue.

○ An image is the graphical and visual representation of some info that can be displayed on computer or printed

What is graphic?

○ Graphics (vector) are elements that are composed of paths.

○ These are resolution independent and can be scaled to any size as they are mathematical description of images.

Graphics
- Bitmap or Raster
- vector

What is raster graphics?

- Raster image may require anti-aliasing which is a technique used in computer graphics to optimize the look of graphics and typography on the display screen.
- It visually smoothes the shape by inserting pixels of intermediate colors along the boundary edges b/w colors.

How to measure the quality of bitmap image?

=> The quality and appearance of a bitmap image is determined by:
- its size
- its resolution

=> "Resolution" is the density of pixels and is expressed in dpi.

=> Bitmap image does not retain their appearance as size increases and so becomes blur.

=> Therefore, bitmap image quality depends on the grid.

=> Larger the grid (resolution) better the quality of image.

- Photoshop, Paintshop pro are bitmap editing application

What are bitmap graphics format?

=> Bitmap graphics format typically describe a rectangular image.

=> The common raster graphics formats are:
- Graphics interchange format (GIF)
- Portable Network graphics (PNG)
- Joint photographic experts group (JPEG)
- Tagged image file format (TIFF)
- Mac PICT (Picture)
- Windows bitmap picture (BMP).

Q# How bitmaps can be inserted?

=> Bitmaps can be inserted by:

(i) By using clip art galleries.

- A clip art gallery is an assortment of graphics, photographs, sound and video related to a single topic.
- Clip arts are popular alternative for users who do not want to create

- own images.
- Clip arts Libraries are available on CD-Roms and on the internet.
- Many graphics applications are shipped with clip art and useful graphics.
- A bitmap image can be download from a website.

+ Procedure.

- Regardless of the source of the image, you should be aware of who owns the copyright to the image & what is required to reproduce the image legally.
    - Legal rights protecting use of images from clips libraries fall into three basic groupings.
    - Public domain images freely used without obtaining permission or licence
    - Royalty free images: are purchased and then used without paying additional license.
    - Rights managed images: - Requires to negotiate with the right holder regarding terms for using the image and how much you will pay for that use.

- Bitmap images can be manipulate, adjust its properties, cut and paste using image-editing program.

ii) Using bitmap Software:-

- The industry standard for bitmap painting and editing programs are:
    - Adobe photoshop and illustrator
    - Macromedia's fireworks
    - Corel's painter
    - Corel draw
    - Quark express.
- The abilities and features of painting and image-editing programs range from simple to complex.

iii). Capturing & editing images:-

- The image on screen is a digital bitmap stored in video memory, updated about every 1/60 of a seconds.
- Another way to assemble images for multimedia by capturing and storing images directly from the screen.

o The PRINT screen button in windows
o Command - Control - Shift - 4 keystroke on the macintosh
- Copies the screen image to the clipboard.
o The simplest way to capture your screen image is press "prope key" from keyboard.
o Image editing programs enable the user to:
o - Enhance and make composite images.
o - Alter and distort images.
o - Add and delete elements
o - Morphing

(IV) Scanning images:

=> Sometimes, having a clip art collections, we still not found ~~the~~ image that is according to our found.
=> For this, every objects can be scanned by using image editing tools.
=> Images can be scanned by users from conventional sources and by making necessary alterations and modifications.

What is morphing?

=> The word morphing comes from the word "metamorphosis" which means a great change.
=> Morphing is a special effect in animations.
=> it transform object shape from one form to another.
=> it is most complicated transformation.
=> it can be used to manipulate still images or to create interesting animated transformation.

What is Photodisc?

=> A commercially available resource of royalty-free images called photodisc, a party of getty images.
=> The photodisc collections contain high-resolution bitmaps with a license for their "unlimited use".

What is monochrom image?

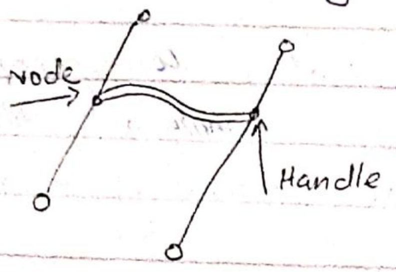
=> Monochrom image just require one bit per pixel, representing black or white.

What are vector-drawn graphics

=> Vector graphics are created from mathematical formulas used to define lines, shapes and curves.

=> Edited in draw program.

=> Shapes can be edited by moving points called nodes (drawing points).



=> Vector graphics are images that are made up of lists of objects and their properties.

=> An object is a mathematically or geometrically defined construct, such as rectangle, line polygon, or circle.

Each of these objects has properties that determine the dimensions (e.g. width, height), appearance (e.g. colour, fill, stroke thickness) and position (e.g. x, y coordinates) of every object in the image. These properties are stored as a list, often called the drawing list.

file format: SVG used to define vector graphics for the web., (.AI), (EPS).

Q. How vector drawing works?

=> A vector is a line that is described by the location of its two endpoints.

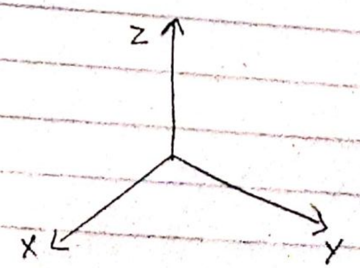
=> Vector drawing uses Cartesian coordinates where a pair of numbers describe a point in 2D space as the intersection of horizontal & vertical lines (the x & y axes).

=> The numbers are always listed in the order x, y.

=> In 3D space, a third dimension - depth - is described by z axis (x, y, z)

Q. In which areas vector-drawn images are used?

- Computer aided design program
- 3D animation programs
- Graphic artists designing for the print media
- Applications requiring drawing of graphic shapes.



Note: vector graphics require plug-in for web based display.

What are Pros & Cons of vector graphics?

Pros	Cons
o Resolution independent	o Lowers color quality
o Small file sizes	o Not good for photographic images.

Popular vector graphic Software:

- o Xara Xtreme
- o Adobe illustrator
- o CorelDraw
- o Inkscape

Q. What is meta graphic?

- o Can Contain vector and raster data.
- o Shapes in vector graphics can be filled with textures and patterns that are raster graphics.
- o Useful when layering text on top of raster graphics.

Examples:

- o WMF – Windows Metafile
- o EPS – Encapsulated postscript

Q. What is difference between bitmaps & vector-drawn images?

Vector-drawn	Bitmap
o Composed of paths.	o Composed of pixels.
o more cost	o Cost less
o Occupy less space in memory because the file size is small.	o Occupy more space which depends on image quality
o Can Convert to Raster	o Can't Convert to vector.
o Scalable	o Loses quality when scaled.
o Easily resizable.	o Not easily resizable.
o Cannot be used for Photorealistic image.	o Can be used for PRI.
o file format is .cdr, .pdf, .dxf, .svg, .eps.	o .jpg, .png, .gif, .tiff, .bmp, .psd
o Software use for - Adobe illustrator - CorelDraw - Figma - Sketch	- Adobe Photoshop - Canva - Digital cameras - Corel painter etc.
Example of use - 2D or 3D animation Logos, Fonts & Laser engravings	- Web imagery - Photography - Print materials

Q What is auto-tracing?

=> The process of connecting a bitmapped image into a vector image is called auto tracing.

=> Bitmapped images, which are represented by dots could have possible grains, halftone dots or other limitations.

=> This can be done by connecting them into vector images.

=> Automatic tracing saves time by using software to detect lines and edges of the bitmap image, and then redraws them as vector images.

Write down possible no of colors for 4-bit, 2-bit & 1-bit?

1-bit - Black & white color (or any two colors)

4-bit - 16 colors

8-bit - 256 colors

16-bit - Thousands of colors

24-bit - more than 16 million colors

What is SVG?

- Scalable vector graphic is an extensible markup language-based vector image format for 2D graphics with support for interactivity & animation.
- SVG format was developed as an open standard format by the W3C.
- The primary use of SVG files are for sharing graphics contents on the internet.

What are Pros and Cons of bitmap image?

Pros

- Precise editing
- Very detailed, photographic images
- multiple tools/filters can be used.
- Can be compressed.
- Widely supported file types.
- Used for scanning & digital photographs

Cons

- Large file size
- when scaled - pixels enlarge and pixels become visible = pixilation
- Device dependent

What are 3D graphics?

=> The three-dimensional transformations are extensions of two 2D transformation.

=> In 2D - two coordinates are used x & y while in 3D - three coordinates x, y and z are used.

⇒ A 3D scene consist of object that in turn contain many small elements, such as blocks, cylinders, spheres, or cones (described in terms of vector graphics).

⇒ The more elements, the finer the objects resolution and smoothness.

⇒ Objects as a whole have properties such as shape, color, texture, shading & location.

⇒ A 3D application lets you model an object's shape, then renders it completely.

Features of 3D application:

1) **Modeling:-**

- Placing all the elements in 3D space is called modeling.
- it involves drawing a shape such as a 2D letter then extruding it or lathing it into a third dimension.
  - extruding:- most straightforward way of making 2D shape into a 3D object is by extruding it.
- An extrusion is simply pushing the 2D shape into a 3D by giving it a Z-axis depth.

Def: The shape of a plan surface extends to some distance is called extrusion.

- Useful for creating block-like shapes, columns, panels etc.
- To create an extruded object, first define the 2D shapes with polylines or splines.
- if create shapes within shapes, such as the two circles, the inner one will create holes in the object.
- Select the desired shapes and apply an extrude operation to them, setting the depth of the object with mouse movement or numerical entity.

  - Lathing:- Rotating a profile of the shape around a defined axis is called lathing.
- a lathed object is a 3D model whose vertex geometry is produced by rotating the points of a spline or other point set around a fixed axis.
- The lathing may be partial; the amount of rotation is not necessarily a full 360 degrees.
  - Used to create 3D symmetrical objects by rotating a 2D shape around an axis. Commonly used in 3D modeling software

- Modeling also deals with lighting, setting a Camera view to project shadows.

2) **Rendering:** → final step in the process of 3D visualizing, which involves creating models of objects, texturing those objects and adding lights to the scene.

When we have completed the modeling of scene or an object in it, then it must be render it for final output.

- Rendering is also called image synthesis.
- it is the process of generating a photo realistic or non-photo realistic image from a 2D or 3D model by means of a Computer program.
- The resulting image is reffered to as the render.
- Use of intricate or Complex algorithms to apply user-specified effects is called rendering.

3) **Shadding:** / Lighting — (Ref on next Page).

⇒ Can be applied in several ways:

flat Shading:- → Technique of lighting
- it is the fastest for the Computer to render and most often used in preview mode.
- Used to shade each Polygon of an object based on direction of light source, their respective color and intensity of light used. To stimulate the differing effect of light and color across the surface of an object intensity.

- Gouraud Shading:- → interpolation used to implement
- Phong shading → Smoothing out multi-surface shapes
- Ray Tracing → To stimulate physical behavior of light

Steps of 3D modeling Process

1) Preparation.
2) Modeling
3) Lighting → Representation of light in Computer generated scene.
   - → Directional light
   - → Area Lighting
   - → Spotlighting
   - → Omnidirectional
4) Rendering

4) **Panorama:**

- A panorama is a long photograph that encompasses a wide area.
- Panorama mode allows you to capture more of the scene by combining images to create a panoramic photo. Also called wide-angle shot or multi-media angle shot.
- Panoramic images are Created by Stitching together a sequence of photos around a circle and adjusting them into a single Seamless bitmap.
- Software such as Ulead Cool 360 and panorama factory are required in order to Create panoramas.
- A Panorama factory is panoramic stitching program for windows & Mac.
- it Creates high quality panoramas from a set of overlapping digital images.

What are 3D animation tools?

- 3D animation, drawing, and rendering tools include:

- Foom*Z
- Google's Sketchup
- Caligari true Space 2
- Autodesk's maya

Q. What is google Sketchup?

- Google Sketchup is 3D modeling software designed for architects, engineers, programmers, and game developers etc.
- Now, it is used to create models.
- it is easier to use.
- Google Sketchup is free, and pay if we want professional version.

Shading refers to the process of altering the color of an object / surface / polygon in the 3D scene, based on thing like the surface angle to light / its distance from lights, its angle to the camera and material properties.

Q. What Pros and Cons of 3D graphics?

Cons

- Complicated overlapping detection
- Needs more advanced programs.
- Requires Costly hardware.
- Requires extensive training.

Pros:

- flexible design
- Rapid prototyping
- Fast design & production
- Cost effective

What is Color?

- Color is a vital component of multimedia.
- Light comes from an atom where an electron passes from a higher to a lower energy level.
- Each atom produces uniquely specific colors.
- Color is the frequency of different light wave within the narrow band of the electromagnetic spectrum, to which the human eye responds.

- Color model is a 3D color ... colors through the primary to produce all range of color through hierarchical system color set. A color model is a ... in which we can create every color this RGB & CMYK

- The letters of the mnemonics ROY G.BIV learned by us to remember many of the color of rainbow, are the ascending frequencies of the visible light spectrum.
- Red, Orange, yellow, green, blue, indigo and violet.
- **Color pickers:** allow us to select a color using one or more different models of color space.

What are two basic methods of making color?

- These are actually two basic methods of making color.
  - ⇒ Additive Color method
  - ⇒ Subtractive color method.

What is additive Color method?

⇒ In additive color method, a color is created by combining colored light sources in three primary colors: Red, green and Blue.

- This is the process used for cathod ray tube, CCD and plasma displays.
- Additive Color starts with black ... colors are at zero intensity they produce black.
- If red and green light are combined at full intensity then they produced yellow light and adds red, green and blue light to produce the visible spectrum of colors.
- As more color is added, the result is lighter when all three colors are combined equally, the result is white light. Uses mixture of light to display colors. The perceived colored depends on the transmission of light. It is used in digital media.
- On digital devices, a red, green or blue element is activated by an electrical charge causing them to glow.
- These elements are called sub-pixels. By combining the three colors, the desired hue is created is one pixel.
- The pixels are then formed like tiny mosaics to create a picture.

What is subtractive Color method?

- In the subtractive color method, color is created by combining colored media such as paint or ink.
- The colored media absorb (or subtract) some parts of the color spectrum of light and reflect the others back to the eye.
- Subtractive color is the process used to create color in printing.

- The subtractive colors are cyan, yellow, magenta and black, also known as **CMYK** (it is four color printing model).
- Paint program uses RGB to create the colors on your monitor.
- While printer uses CMYK to print out images.
- Subtractive colors begin with white & ends with black, as color is added the result is darker.
- Adding equal amounts of cyan, yellow and magenta will produce black, but in reality, the result is often a very muddy dark brown.
- To produce a true black, black pigment is added. Black referred to as "k", or the key color, and is also used to add density.

When to Choose RGB VS CMYK?

- It is important to choose the correct color methods at the beginning of the project to get the best results.
- If the final product is for print, then convert the color mode from RGB to CMYK but if the final product only appears on a screen or monitor, keep the color mode as RGB.
- Because additive colors use transmitted light, the colors appear much brighter and create a larger visible spectrum, producing millions of colors on a screen.
- Subtractive colors use reflected light, so they appear muted in contrast.

What are Computer Color models:

- Models or methodologies used to specify colors in computer terms are:
  - RGB
  - HSL
  - CMYK
  - CIE

RGB model:

- It is 24-bit methodology.
- Color is specified in terms of red, green and blue values ranging from 0 to 255.
- Eight bits of memory are required to define these 256 possible choices, and that has to be done for each of the three primary colors. A total of 24-bits of memory $8+8+8=24$ are needed to describe the exact color.

which is one of "millions" $256 \times 256 \times 256 = 16,777,216$).

- Rather than using one numbers between 0 and 255, two hexadecimal numbers use $16 \times 16 = 256$ bit.

HSB and HSL Color model:-

- HSB stands for hue, saturation, brightness and HSL stands for hue, saturation, lightness.
- Colors are specified in this model as an angle 0 to 360 degrees on a color wheel, and saturation, brightness, and lightness as percentages.
- Saturation is the intensity of a color. At 100 percent saturation a color is pure, at 0 percent saturation, the color is white, black or gray.
- Lightness or brightness is the percentage of black or white that is mixed with a color.
- A lightness of 100% will yield a white color, 0 percent is black, the pure color has a 50% lightness.

CMYK model:

- Not applicable for multi-media production.

CIE model:

- Describe color values in terms of frequency, saturation and illuminance.
- it specifies color composition as function of x (red) and y (green).
- For any value of x and y, value of z (blue) can be found as

$$z = 1 - (x+y)$$

- The (x,y,z) values of a color specifies percentage of red, green and blue needed to form the color (Trichromatic co-efficients).

CMY/CMYK models:

-
$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Pros and Cons of RGB model:

Pros:

- Straightforward
- Matches well with human vision systems strong response to Red, green and blue.

Cons:

- Difficult for human description of color.
- Highly redundant and correlated.

HSI color model:

- Useful for human color interpretation.
- Three axis represent:

Hue: - represent dominant wavelength of the color, typically measured in degrees. it defines the, [illegible] color RGB.

- Describes pure/dominate color perceived by observer

Saturation: (Purity of color) or intensity of the color

- Amount of white light mixed with hue.
- High saturation = hight purity = little white light mixed with hue.
- A saturation value [illegible]

Intensity: - indicates the perceived intensity of the color

- Brightness.
- A brightness value of 0 results in black, while a value 1 represent full brightness.

YCC model:-

- it is used for photo CD images and has been developed by kodak.
- it represents the three RGB channels (Y) and two color difference channels, CO

YIQ and YUV models:

- Two common color models in video are YUV and YIQ.
- YUV uses properties of the human eye to prioritize information.
- Y is the black and white (luminance) image, U and V are the color difference (chrominance) images.
- Used in television broadcasting.

- it represents color by separating:

=> Y (luminance)

Cb and Cr are two chrominance components

Represent the brightness or intensity of the color.

- it ranges from 0 (black) to 1 (white)
- it is derived from the RGB color model but emphasize the brightness information.

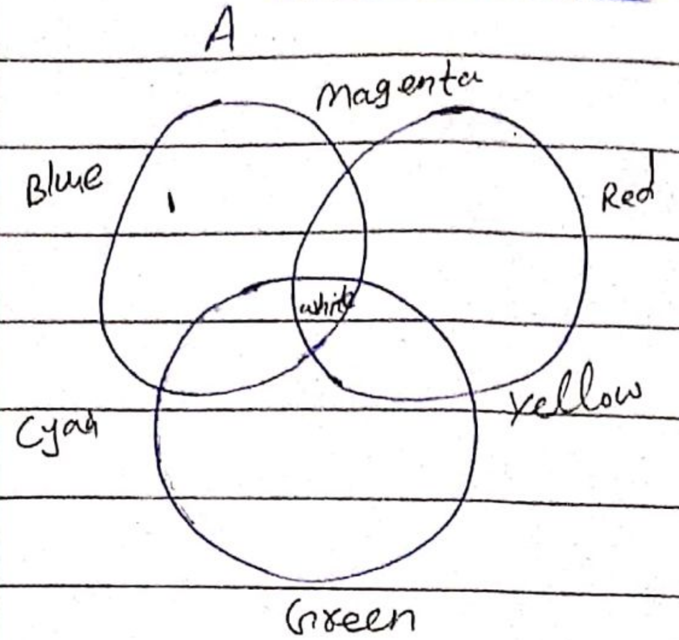
-> Capture the color information and represent the differences b/w red and Y and blue and Y respectively. range from - to + value.

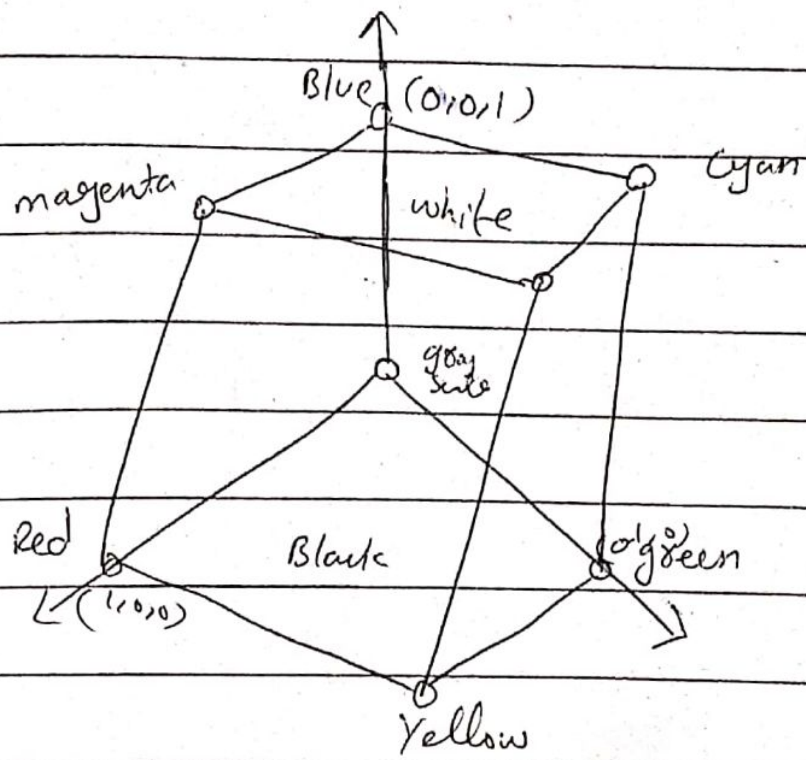
- Commonly used in video & image compression techniques such as jpeg and mpeg.

M T W T F S

تاریخ: ___/___/___

Additive color





Example:

=> Color monitor

=> Television

etc.

video are

Cons:

- Di
- Hi (

HSI

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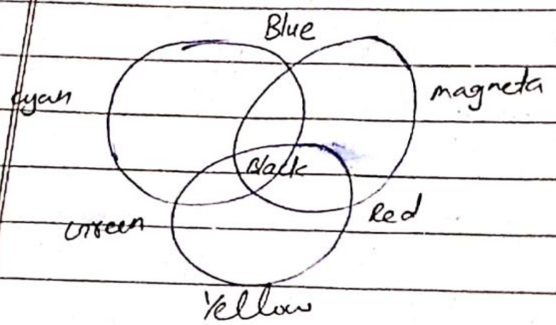
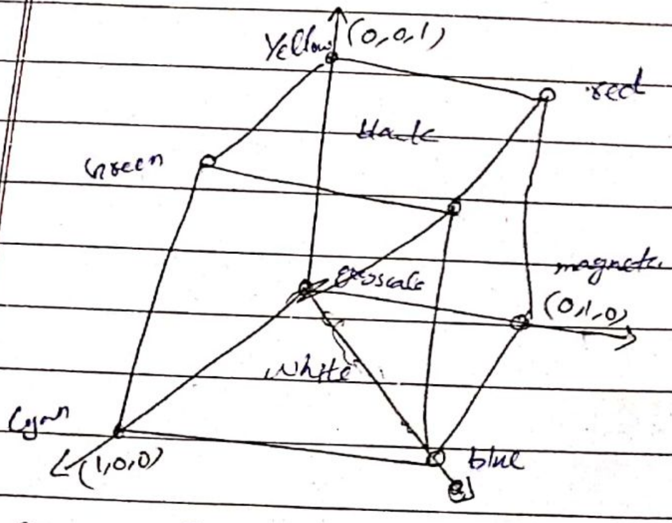
and two Co...

Subtractive color





ex: Paint

Pigments

Color filter

Pros:

- Easy to implement
- uses color space for application.
- No transformation for data display.

Cons

- we cannot transfer the color values from one to another device.
- Complex to determine the particular color.

Color look-up table:

- Technique or process to Convert:-
- a range of input Colors into another range of o/p colors.
- it existed in the graphics card.
- Graphics card is also called display ...

M T W T F S تاریخ: __/__/__

Color palettes:-

=> Mathematical tables used to determine the color of the pixel that is displayed on the screen.

=> In macintosh OS; it is known as "color look up table".

=> In windows OS, it is known as the "color palette."

=> It is used to stores a set of bytes instead of the color of the image.

Can be generated manually by designers or automatically by computer algorithms.

Q. What are color plettes?

- Palettes are mathematical tables that define the color of pixels displayed on the screen. Predefined set of colors.
- Palettes are called "Color lookup tables" or CLUTs on the Macintosh.
- The most common palettes are 1, 4, 8, 16 and 24-bit deep.
- Paint programs provide a palettes tool for displaying available colors
- Color graphics adaptors work with 256 shades of each color producing over 16 million colors (256*256*256).

Q. What is palettes flashing problem?

- Palette flashing occurs when you use a series of images each with its own color palette.
- when the new image replaces the older one a flash occurs on the screen - a serious problem in multimedia.
- Solution:
  - Use a single palette for all project images or
  - fade each image to white or black before showing the next image since white and black are present in most palettes.

(digital halftoning)

What is dithering?

Types
- Random dither
- Ordered dither
- Error diffusion dith

- Dithering is a process where the color value of each pixel is changes to the closest matching color value in the target palette, using a mathematical algorithm.
- It "averages" the color over an area and is usually close to the original color. - used to describe the strategic application of noise to an image.
- Dithering Software is usually built into image editing and multi-media programs. Traditionally used to improve the appearance of images whose color is limited to a particular set.

(Long)

Q What are the image file formats? Describe it?

File format:

- A file format is the structure of how information is stored in a computer file.
- File formats are designed to store specific types of information.

Categories of files:

- Uncompressed
- Lossless
- Lossy

Common file formats used in multi-media are as follows:

Macintosh formats:-

=> On the macintosh, the most commonly used format is PICT.— are encoded in QuickDraw Command.

=> PICT is a Complicated and versatile format developed by Apple. in 1984.

=> Almost every image application on the macintosh can import or export PICT files.

=> In a PICT file, both vector drawn objects and bitmaps can reside side-by-side.

=> PICT stands for personalized image capture technology.

=> The format is used for interchanging bitmap and vector graphics b/w macintosh applications.

Versions:
- PICT 1
- PICT 2

=> it is meta-format.

Algorithm use: JPEG, Packbits

Windows format:-

=> The most commonly used image file format on windows is DIB, also known as BMP.

=> DIB stands for device-independent bitmaps.

o Bitmap formats used most often by windows developers are:

- BMP
- TIFF
- PCX

- Color depth range (1-bit to 24-bit)

1) BMP: (.bmp)

=> Bitmap image file format is developed by microsoft. - it can store data as 2D digital images in monochrome as well as in color format.

=> There is no compression or information loss with BMP files which allow images to have very high quality, but also very large file sizes. Donot support transparency & animation.

Compression: None

Best for: high quality Scans, archival Copies.

Algorithm: Huffman and RLE

2) TIFF: (.tif, .tiff)

Support both compressed & uncompressed.

=> Tagged image file format are lossless images files. - can store metadata such as image resolution, color space etc.

=> it store rasters images.

=> Do not lose any image quality or information. Created by Aldus Corporation, in 1986.

- store multiple images per file

Compression:

- Lossless - no Compression. very high quality images.

Special attributes:

- Can Save transparencies

Pros
- good quality image
- contain limited colour palette 256

Cons
- file size is large
- compatible only PC.

Best for high quality prints, professional publications, archival copies.

(3) **PCX**: Developed by Zsoft corporation in 1980s
- lossless

=> PCX Stands for picture exchange, is a raster image file Type.

Algorithm:
- RLE

=> it uses colour pixels to display images.

=> Capable of storing only a single image per file. - contain 256 color, various resolutions
320 X 200 X
640 X 480

=> Used by MS-DOS paint Software.
- Not widely used today

Cross-platform formats:
- replaced by JPEG & Png

=> JPeg. gif. Png most commonly used format on the web.

(1) JPeg:

- Stands for joint photographic experts groups.
- it is a 'lossy' format, means the image is compressed to make a smaller file. it retain high quality image quality.
- The Compression does create a loss in quality but this loss is generally not noticeable.
- it is most popular on internet and for digital cameras.

Compression:
- Some information is lost.

Best for:
- web images, non-professionals printing. E-mail, powerpoint.

Special attributes:
- Can choose amount of compression when saving in image editing programs like Adobe photoshop.

(b) Gif (.gif):
- Develop in 1987
- -> support both animated & static images. Great for webpage banners, ads

- Graphics interchange format files widely used for web graphics, because they are limited to only 256 colors, can allow for transparency, and can be animated.

Gif files are typically small in size and are very portable.
- Stephen Ewl Wilhite.

Compression:
- Compression without loss of quality.
Lossless

Best for: - web images

Algorithm
- LZW
Lempel
Ziv
Welch

Special attributes:
- Can be animated
- Can save transparency

(c) PNG (.Png):
- Stands for portable network graphics files.

- To overcome the limitation of GIF
- it allows you include alpha channel.
- it is a lossless image format originally designed to improve upon and replace the gif format.
- PNG files are able to handle up to 16 million colors, unlike the 256 colors supported by Gif.

Compression:

- Lossless - Compression without loss of quality.
- Best for: web images
- Special attributes:
  - Save transparency
- One drawback is large file size which can be mitigated using tools without loss of quality.

4) EPS (.eps): → Created by Adobe. Use mathematical curves.
- An encapsulated Postscript file is a common vector file type. Standard format for print industry.
- Can be opened in many illustration applications such as adobe illustrator or Coreldraw.

Compression: None Best for: illustrations, vector artwork

Special attributes: Saves vector info

5) PDF (1990s) → used to represent document in a way that is independent of hardware / sw and os. Used to create, view
- Portable document format manages multi-media content. Created by Adobe
- PSD, AI, CDR, DXF - proprietary formats used by applications.

Cons: Read only. - Great vector image format.